Georg Wölflein

Researcher in deep learning for medicine with seven years of industry experience in computer vision and a strong software engineering background. Passionate about both theory and practice, my research focuses on LLM agents for medicine as well as computer vision for computational pathology.

Education

Aug 2021 – Sep 2025 **Doctor of Philosophy (PhD), Computer Science**, *University of St Andrews*, Scotland.
During my PhD, I developed novel deep learning models for computational pathology, as well as an agentic framework enabling LLMs to autonomously create research tools in medicine and life sciences, resulting in first-author publications at ECCV, ACL, WACV, and others. My PhD was supervised by Dr Ognjen Arandjelović.

Sep 2017 – Jun 2021 (2nd year direct entry) **Master in Science (MSci), Computer Science**, *University of St Andrews*, Scotland.
First-Class Honours, GPA: 95%
Master's thesis: "Determining chess game state from an image" (grade: 20.0/20).
Honours level courses include machine learning, AI principles & practice, language & computation, data-intensive systems, information visualisation, concurrency & multi-core architectures, constraint programming, software architecture, software engineering, complexity, OS, databases, data encoding, component technology, logic, software verification, compiler design & implementation.

2005 – 2017 **International Baccalaureate and Abitur**, *Dresden International School*, Germany.
IB Diploma: 40 points, German Abitur: 1.3
Valedictorian. Higher level subjects: maths, physics, computer science.

Experience

Jun 2025 – present **AI Engineer**, *Synagen AI*, Dresden, Germany
Developing LLM agents for medicine.

Nov 2024 – May 2025 **Researcher**, *Else Kröner Fresenius Center for Digital Health (EKFZ)*, Dresden, Germany
At Kather Lab, I conducted research on LLM agents in medicine and deep learning for computational pathology. Specifically, I developed ToolMaker, an LLM agent that autonomously creates research tools from research papers with code repositories (accepted at ACL 2025). I also supervised multiple PhD students (resulting in publications at CVPR & MICCAI), led an LLM agents reading group, and managed the team developing our open-source computational pathology platform STAMP.

Jul-Oct 2024 **Research Intern**, *Google*, London, UK
Statistically evaluated FitBit's activity recognition model across diverse activity classes and demographic groups. Identified and quantified key failure modes, performed an in-depth analysis of root causes, and communicated actionable recommendations for model improvement to cross-functional teams.

Jul-Sep 2023 **Research Intern**, *Else Kröner Fresenius Center for Digital Health (EKFZ)*, Dresden, Germany
At Kather Lab, I performed an extensive benchmarking study of self-supervised feature extractors in histopathology (accepted at ECCV), and contributed to various LLM-related projects.

Jan-Mar 2023 **PhD Placement**, *Lay Summaries Ltd*, Glasgow, Scotland
Developed a NLP pipeline for automating the generation of lay summaries of EU clinical trials in a part-time PhD placement funded by The Data Lab.

May 2018 – May 2022 **Working Student – Computer Vision**, *Robotron Datenbank-Software*, Dresden, Germany
Gained practical experience in deep learning and software engineering by developing deep learning models and deploying them to production in the Realtime Computer Vision (RCV) department.
- Designed and implemented containerised infrastructure for automatically training, evaluating, and deploying TensorFlow and PyTorch models for industrial use cases.
- Selected and trained deep learning models for various industrial use cases, including a system for a car manufacturer that reduced the error rate of detecting faulty parts by 90%.

Jun-Aug 2019 **Software Engineering Intern**, *J.P. Morgan*, Glasgow, Scotland
Developed a data visualisation and reporting dashboard using Python, React, TypeScript, SQL. Gained hands-on experience with Scrum, teamwork, and prioritising requirements from different stakeholders.

Skills

Programming Python, C/C++, Java, SQL, JavaScript, TypeScript, Haskell, C#, LaTeX

georg@woelflein.de • *georg.woelflein.eu* • *georg-wolflein* • *georg-wolflein*
scholar

Technologies	PyTorch, Triton, TensorFlow, Keras, JAX, Docker, Dask, Apache Spark, Postgres, React
Languages	German, English *(mother tongue)*; French *(B1)*

Prizes and awards

2020	Adobe Prize (£750) for the highest GPA in Senior Honours Computer Science
2018 – 2021	4x Dean's List Award of Academic Excellence at the University of St Andrews
2017	Valedictorian at Dresden International School; subject awards for maths and computer science
2010 – 2017	12x High Honour Roll (GPA over 6.0 of 7) at Dresden International School

Publications

Conference papers

2025 M. Ligero, T. Lenz, **G. Wölflein**, O. S. M. El Nahhas, D. Truhn, and J. N. Kather, "Abnormality-driven representation learning for radiology imaging," in *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Sep. 2025. (link)

G. Wölflein, D. Ferber, D. Truhn, O. Arandjelović, and J. N. Kather, "LLM agents making agent tools," in *Proceedings of the Annual Meeting of the Association for Computational Linguistics (ACL)*, Jul. 2025. (link)

T. Lenz, P. Neidlinger, M. Ligero, **G. Wölflein**, M. van Treeck, and J. N. Kather, "Unsupervised foundation model-agnostic slide-level representation learning," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, Jun. 2025, pp. 30 807–30 817. (link)

2024 O. S. M. El Nahhas, **G. Wölflein**, M. Ligero, T. Lenz, M. van Treeck, F. Khader, D. Truhn, and J. N. Kather, "Joint multi-task learning improves weakly-supervised biomarker prediction in computational pathology," in *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, Springer Nature Switzerland, Oct. 2024, pp. 254–262. (link)

G. Wölflein, D. Ferber, A. R. Meneghetti, O. S. M. El Nahhas, D. Truhn, Z. I. Carrero, D. J. Harrison, O. Arandjelović, and J. N. Kather, "A good feature extractor is all you need for weakly supervised pathology slide classification," in *European Conference on Computer Vision (ECCV)*, BioImage Computing Workshop (oral), Springer, Sep. 2024. (link)

2023 **G. Wölflein**, I. H. Um, D. J. Harrison, and O. Arandjelović, "HoechstGAN: Virtual lymphocyte staining using generative adversarial networks," in *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, Jan. 2023, pp. 4997–5007. (link)

Journal articles

2026 Y. Liu, Z. I. Carrero, X. Jiang, D. Ferber, **G. Wölflein**, L. Zhang, S. Jayabalan, T. Lenz, Z. Hui, and J. N. Kather, "Benchmarking large language model-based agent systems for clinical decision tasks," *npj Digital Medicine*, vol. 9, no. 1, p. 259, Feb. 2026. (link)

2025 D. Ferber, O. S. M. El Nahhas, **G. Wölflein**, I. C. Wiest, J. Clusmann, M.-E. Leßman, S. Foersch, J. Lammert, M. Tschochohei, D. Jäger, M. Salto-Tellez, N. Schultz, D. Truhn, and J. N. Kather, "Development and validation of an autonomous artificial intelligence agent for clinical decision-making in oncology," *Nature Cancer*, vol. 6, no. 8, pp. 1337–1349, Jun. 2025. (link)

2024 D. Ferber, **G. Wölflein**, I. C. Wiest, M. Ligero, S. Sainath, N. G. Laleh, O. S. M. El Nahhas, G. Müller-Franzes, D. Jäger, D. Truhn, and J. N. Kather, "In-context learning enables multimodal large language models to classify cancer pathology images," *Nature Communications*, vol. 15, no. 1, p. 10 104, Nov. 2024. (link)

*equal contribution

O. S. M. El Nahhas, M. van Treeck, **G. Wölflein**, M. Unger, M. Ligero, T. Lenz, S. J. Wagner, K. J. Hewitt, F. Khader, S. Foersch, D. Truhn, and J. N. Kather, "From whole-slide image to biomarker prediction: End-to-end weakly supervised deep learning in computational pathology," *Nature Protocols*, vol. 20, no. 1, pp. 293–316, Sep. 2024. (link)

D. Ferber, I. C. Wiest, **G. Wölflein**, M. P. Ebert, G. Beutel, J.-N. Eckardt, D. Truhn, C. Springfeld, D. Jäger, and J. N. Kather, "GPT-4 for information retrieval and comparison of medical oncology guidelines," *NEJM AI*, vol. 1, no. 6, AIcs2300235, May 2024. (link)

2023 **G. Wölflein***, I. H. Um*, D. J. Harrison, and O. Arandjelović, "Whole-slide images and patches of clear cell renal cell carcinoma tissue sections counterstained with Hoechst 33342, CD3, and CD8 using multiple immunofluorescence," *Data*, vol. 8, no. 2, Feb. 2023. (link)

2022 R. De Filippis*, **G. Wölflein***, I. H. Um, P. D. Caie, S. Warren, A. White, E. Suen, E. To, O. Arandjelović, and D. J. Harrison, "Use of high-plex data reveals novel insights into the tumour microenvironment of clear cell renal cell carcinoma," *Cancers*, vol. 14, no. 21, Nov. 2022. (link)

2021 **G. Wölflein** and O. Arandjelović, "Determining chess game state from an image," *Journal of Imaging*, vol. 7, no. 6, Jun. 2021. (link)

Preprints

2024 D. Ferber, L. Hilgers, I. C. Wiest, M.-E. Leßmann, J. Clusmann, P. Neidlinger, J. Zhu, **G. Wölflein**, J. Lammert, M. Tschochohei, H. Böhme, D. Jäger, M. Aldea, D. Truhn, C. Höper, and J. N. Kather, *End-to-end clinical trial matching with large language models*, under review, Jul. 2024. (link)

G. Wölflein, D. Ferber, A. R. Meneghetti, O. S. M. El Nahhas, D. Truhn, Z. I. Carrero, D. J. Harrison, O. Arandjelović, and J. N. Kather, *Benchmarking pathology feature extractors for whole slide image classification*, Jun. 2024. (link)

2023 **G. Wölflein**, L. C. Magister, P. Liò, D. J. Harrison, and O. Arandjelović, *Deep multiple instance learning with distance-aware self-attention*, May 2023. (link)

Abstracts

2025 L. Zhang, **G. Wölflein**, D. Ferber, J. Liang, Z. I. Carrero, T. Lenz, and J. N. Kather, *74P self-hosted open-source large language models for autonomous clinical agents*, ESMO Real World Data and Digital Oncology, Nov. 2025. (link)

2023 D. Alouges, **G. Wölflein**, I. H. Um, D. Harrison, O. Arandjelović, C. Battail, and S. Gazut, *Performance comparison between federated and centralized learning with a deep learning model on Hoechst stained images*, ISMB/ECCB Abstracts, Jul. 2023. (link)

Datasets

2022 **G. Wölflein**, I. H. Um, D. J. Harrison, and O. Arandjelović, *Whole slide images and patches of clear cell renal cell carcinoma counterstained with multiple immunofluorescence for Hoechst, CD3, and CD8*, BioImage Archive, Dec. 2022. (link)

2021 **G. Wölflein** and O. Arandjelović, *Dataset of rendered chess game state images*, Open Science Foundation, May 2021. (link)

Invited talks

Sep 2025 **AI in Medicine Summer School 2025, TU Dresden**
LLM agents for biomedical research

Mar 2025 **Center for Scalable Data Analytics and Artificial Intelligence (SCADS.AI), Dresden**
LLM agents for autonomously creating research tools in medicine and life sciences (link)

Feb 2024 **Tissue Image Analytics (TIA) Centre, University of Warwick**
A good feature extractor is all you need for weakly supervised learning in histopathology (link, video)

Apr 2022 **KATY EU Project**
High-plex data reveal novel insights into the tumour microenvironment of clear cell renal cell carcinoma (with Raffaele De Filippis)

Courses and training

Dec 2023 Focus on Peer Review, *Nature Masterclasses*.

Mar 2023 Clinicum Digitale, *Dresden University of Technology*: attended a two-week interdisciplinary spring school on medicine and technology.

Jun 2020 Deep Learning Specialisation, *Coursera*.

May 2020 PyTorch for Deep Learning and Computer Vision, *Udemy*.

Sep 2019 Mathematics for Machine Learning Specialisation, *Coursera*.

Sep 2019 TensorFlow 2.0: A Complete Guide on the Brand New TensorFlow, *Udemy*.

2013 – 2014 C/C++ Course, *Volkshochschule Dresden (Community College Dresden)*.

Service

2025 Reviewer for *MICCAI*, *MICCAI COMPAYL*

2024 Reviewer for *ECCV BIC*

2023 Reviewer for *CVPR* and *npj Precision Oncology*

Volunteering

2021 – 2022 **Developmental squad representive**, *University of St Andrews Volleyball Club*
I was elected to represent the recreational volleyball team within the committee and club decision-making process. This role helped me improve communication and organisational skills.

2018 – 2020 **Secretary**, *University of St Andrews Muscle and Athletics Sports Society (MASS)*
As secretary of MASS, I was in charge of coordinating meetings, writing minutes, and taking care of administrative tasks. This position has helped me develop teamwork and organisational skills.

2010 – 2017 **Volunteer firefighter**, *Freiwillige Feuerwehr Possendorf*
I am passionate about giving back to the community. Since age eleven, I was a youth fire fighter in my local fire department. In 2015, I completed the training qualification to become a member of the adult fire department, and participated in active service until I moved to Scotland in September 2017.

In my free time, I enjoy playing chess, volleyball, lifting weights, and improvising on the piano.